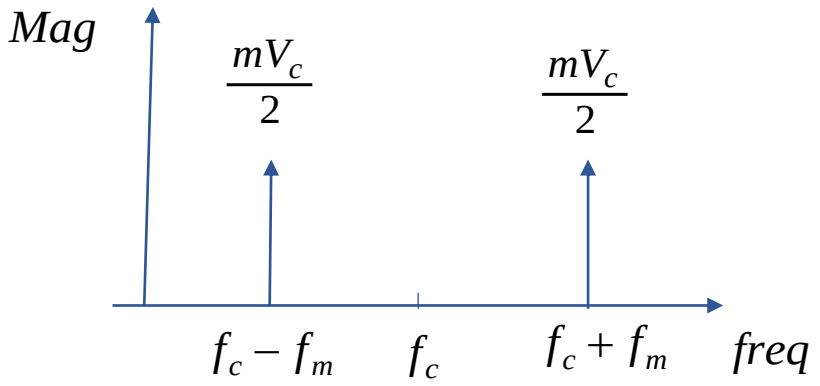
DSBSC-AM

- For Double sideband with suppressed carrier (DSBSC) AM
  - $k = 1$, $C = 0$ and $\phi_c = 0$

$$V_{am}(t) = V_m(t)\cos\omega_c t$$
$$= \frac{mV_c}{2}\cos(\omega_c + \omega_m)t + \frac{mV_c}{2}\cos(\omega_c - \omega_m)t$$



DSBSC-AM …/2

- Power generated in a resistance R

$$P_{am} = \frac{m^2 V_c^2}{4R} = \frac{m^2}{2} P_c = P_{SB}$$

DSBSC-AM …/3

Exercise 2

- Repeat exercise 1 for a DSB-SC system and compare your results with those for the DSBC case.

Example

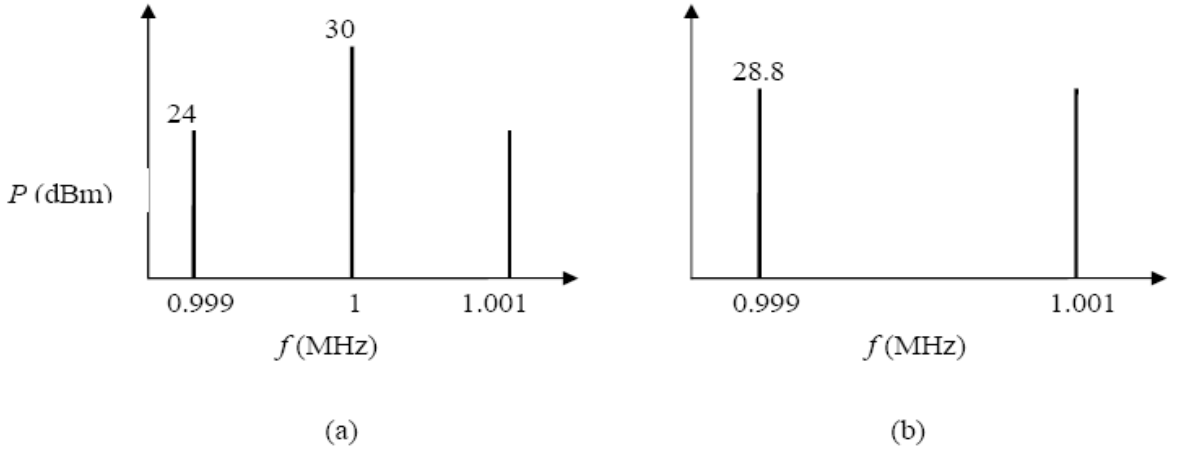
- effect of removing the carrier from a fully modulated AM signal in the frequency domain

$(a)\ f_c = 1MHz, \quad f_m = 1kHz,$

$P_c = 1W = 30\,dBm$

$P_{LSB} = P_{USB} = 0.25W = 24\,dBm$

$P_{am} = P_c + P_{SB} = 1 + 0.5W = 1.5W$



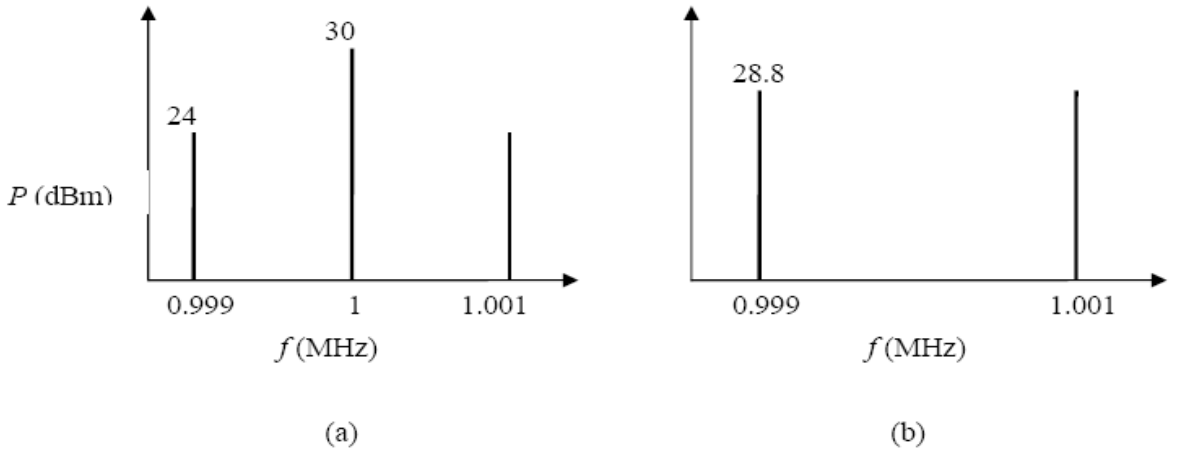
DSBSC-AM …/4

Example

- effect of removing the carrier from a fully modulated AM signal in the frequency domain

$(b)\ P_{SB} = P_{am} = 1.5W$

$P_{LSB} = P_{USB} = 0.75W = 28.8\,dBm$



SSB-SC-AM

- USB & LSB contain the same information.

- Suppressing f_c & eliminating 1 sideband gives SSB

$$V_{am}(t) = \frac{mV_c}{2}\cos(\omega_c + \omega_m)t \quad or \quad \frac{mV_c}{2}\cos(\omega_c - \omega_m)t$$

- SSB bandwidth

$$B_{SSB} = \frac{1}{2}B \text{ of } DSBC \quad or \quad DSB - SC$$

SSB-SC-AM ../2

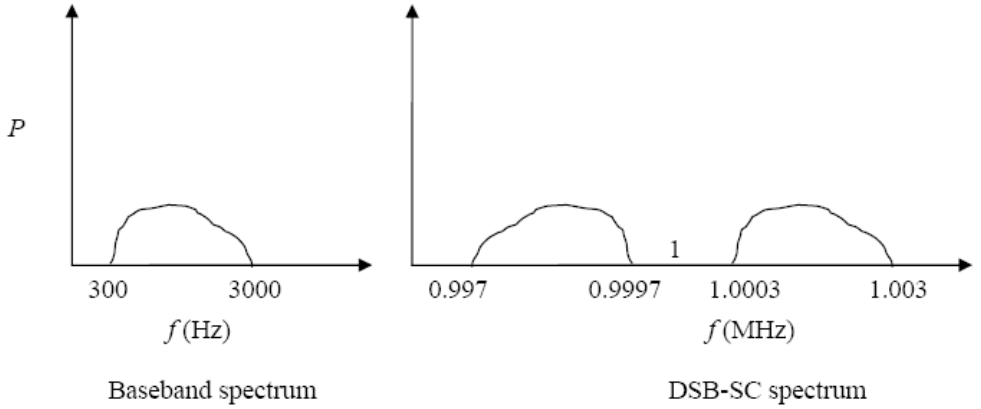
- Bandwidth improvement will be > 2

$f_m = 300\,Hz \text{ to } 3\,kHz; \qquad f_c = 1\,MHz$

$B_{DSBSC} = 2 f_{m_{\max}} = 6\,kHz$

$B_{SSB} = B \text{ of one sideband}$

$= 3kHz - 0.3\,kHz = 2.7\,kHz$



SSB-SC-AM ../3

Benefits of bandwidth (B) reduction?

Power in SSB-SC-AM ../4

Power generated in R Ω by the SSB-SC modulated signal is:

$$P_{am} = \frac{m^2 V_c^2}{8R} = \frac{m^2}{4} P_c = \frac{1}{2} P_{SB}$$

Exercise 3

- Repeat exercise 1 for an SSB-SC system and compare your results with those for the DSBC and DSB-SC cases.

Vestigial Sideband (VSB) - AM

VSB

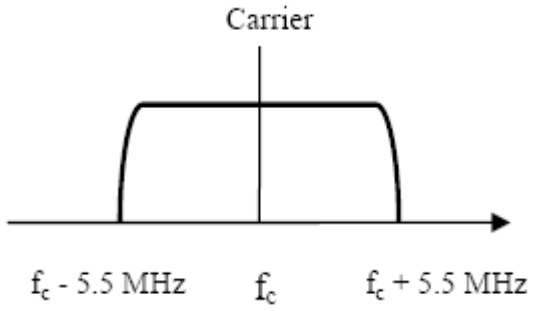
- is used for wideband signals, such as television signal

- DSBC is not suitable for use with wideband signals?

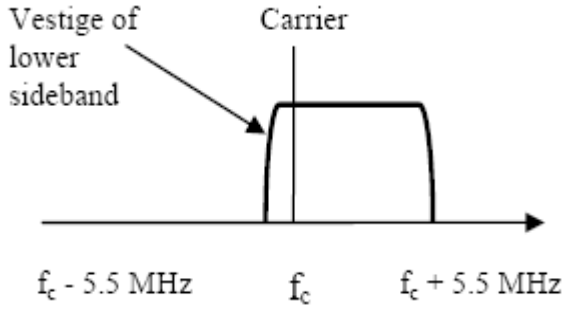
Vestigial Sideband (VSB) - AM

- It would be extremely difficult to produce SSB-AM for a television signal?

- Part of one sideband and the whole of the other is transmitted



DSB AM Transmission

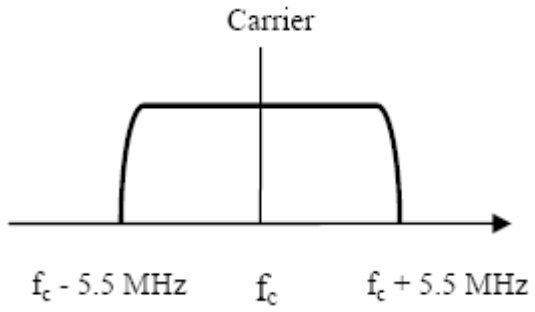


VSB AM Transmission

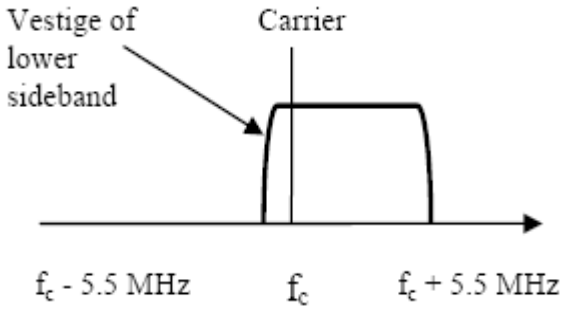
Vestigial Sideband (VSB) - AM

- The VSB-AM signal compared to
  - DSBC / DSB-SC AM
  - SSB-AM

- Benefit of using VSB-AM signal…?



DSB AM Transmission



VSB AM Transmission